TP6 :Opérations morphologiques

Telecharger : smarties.png

Exemple 1 (avec skimage)

Eléments structurants

```
import numpy as np
from skimage import data
import matplotlib as plt
from skimage import morphology
from matplotlib.pyplot import imshow, get_cmap

imshow(morphology.disk(1), cmap=get_cmap('gray'))
```

Image(avant dilatation)

```
image_test = np.array([[0,0,0,0,0],
                [0,1,0,0,0],
                [0,0,0,0,0],
                [0,0,0,1,0],
                [0,0,0,0,0],
                [0,1,0,0,0]])
imshow(image_test, cmap=get_cmap('gray'))
```

Dilatation

```
dilation = morphology.binary_dilation(image=image_test,
                          selem=morphology.disk(1))
imshow(dilation, cmap=get_cmap('gray'))
```

image avant l'erosion

```
image_test = np.array([[0,0,0,0,0],
                [0,0,1,0,0],
                [0,1,1,1,0],
                [0,0,1,0,0],
                [0,0,0,0,0]])
imshow(image_test, cmap=get_cmap('gray'))
```

Erosion

```
erosion = morphology.binary_erosion(image_test, morphology.disk(1))
imshow(erosion, cmap=get_cmap('gray'))
```

Question : remplacer morphology.disk(1), par morphology.disk(2) et appliquer la dilatation et l'érosion

Exemple 2 (avec opencv)

```
import cv2
```

```
import numpy as np
from matplotlib import pyplot as plt
img = cv2.imread('smarties.png', cv2.IMREAD_GRAYSCALE)
#convertir l'image en noir et blanc
_, img_nb = cv2.threshold(img, 220, 255, cv2.THRESH_BINARY_INV)
#element structurants
kernal = np.ones((2,2), np.uint8)
dilation = cv2.dilate(img_nb, kernal, iterations=2)
erosion = cv2.erode(img_nb, kernal, iterations=1)
opening = cv2.morphologyEx(img_nb, cv2.MORPH_OPEN, kernal)
closing = cv2.morphologyEx(img_nb, cv2.MORPH_CLOSE, kernal)
```

Afficher les résultats

```
cv2.imshow('dilit',dilation)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Question : remplacer kernel par np.ones((3,3), np.uint8)

Afficher les résultats sur un plots

```
titles = ['image','img_noir_blanc', 'dilation', 'erosion', 'opening', 'closing']
images = [img,img_nb, dilation, erosion, opening, closing]
for i in range(6):

    plt.subplot(2, 4, i+1), plt.imshow(images[i], 'gray')

    plt.title(titles[i])
```

Question : remplacer kernel par np.ones((3,3), np.uint8)